Exercises from Lectures 1–3. Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified.

Exercise 1. Projectives, endomorphisms, and duals over $\Bbbk A_3$

(1) For $A = \Bbbk A_n$ with $n \geq 2$, verify that $A_d = \operatorname{span}_{\Bbbk}\{e_{ij} \mid j - i = d\}$ defines a grading on A

(2) Compute $\operatorname{rad}(\Bbbk A_n)$ from its definition as the intersection of all maximal right ideals.

(3) Let $A = \Bbbk A_3$, $M = e_{22}A$, and consider $\mathbb{D}(M)_d = \operatorname{Hom}_{\Bbbk}(M_{-d}, \Bbbk)$ with $(af)(m) = f(ma)$.

i. Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on it.

ii. Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that $\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$ as graded left A-modules.

iii. Deduce that $\mathbb{D}(M)$ is not isomorphic to any $(Ae_{jj})(r)$, $j \in \{1, 2, 3\}$, $r \in \mathbb{Z}$.

(4)

Exercise 2. Dependence on the symmetrising form
Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, $V = \Bbbk e_{12}$, and consider $M = e_{11}A$, with inherited gradings. For $c = (c_1, c_2) \in (\Bbbk \setminus \{0\})^2$ define

$$\tau_c : S \to \Bbbk, \qquad ae_{11} + be_{22} \mapsto c_1 a + c_2 b.$$

Recall that $M^{(*)} = \bigoplus_d \operatorname{Hom}_S(M_{-d}, S)$ with the displayed summand in degree d, that $\alpha_c : M^{(*)} \to \mathbb{D}(M)$ is $f \mapsto \tau_c \circ f$, and that

$$a \cdot_c f = \alpha_c^{-1}(a\, \alpha_c(f)), \qquad (a\lambda)(m) = \lambda(ma).$$

(1) Verify that τ_c is symmetrising. Let $\varepsilon : V \to \Bbbk$ be $\varepsilon(ue_{12}) = u$. Find the right S-linear map $f : V \to S$ and the left S-linear map $g : V \to S$ with $\tau_c \circ f = \varepsilon = \tau_c \circ g$, and determine whether f is also left linear and g also right linear .

(2) Let $\varphi_0(ae_{11} + be_{12}) = ae_{11}$ and $\varphi_1(ae_{11} + be_{12}) = be_{22}$. Show that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$ and find their degrees.

(3) Compute $e_{11} \cdot_c \varphi_i$, $e_{22} \cdot_c \varphi_i$, $e_{12} \cdot_c \varphi_i$ for general c and $i \in \{0, 1\}$. For c, c', prove that $\cdot_c = \cdot_{c'}$ if and only if (c_1', c_2') is a scalar multiple of (c_1, c_2).

(4) i. Explain, without computing any action, why $(M^{(*)}, \cdot_c) \cong (M^{(*)}, \cdot_{c'})$ as graded left A-modules for *all* c, c'.
ii. Write the isomorphism $t = \alpha_{c'}^{-1}\alpha_c$ explicitly on φ_0, φ_1, and check directly that it commutes with e_{11}, e_{22}, e_{12}.
iii. Take $c = (1, 1)$ and $c' = (1, 2)$, $\operatorname{char} \Bbbk \neq 2$: what is t?

Exercise 3. Tensor products and S-duals
Let S be a finite-dimensional semisimple $\Bbbk$-algebra. Let V be a finite-dimensional right S-module and W a finite-dimensional S-bimodule, and define

$$\theta_r : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*, \qquad \theta_r(g \otimes f)(v \otimes w) = g(f(v)w).$$

(1) Prove that θ_r is well defined: check bilinearity in (g, f) and in (v, w), invariance under $vs \otimes w \leftrightarrow v \otimes sw$ and $gs \otimes f \leftrightarrow g \otimes sf$, and right S-linearity of $\theta_r(g \otimes f)$. When V is a bimodule, show $\theta_r(sg \otimes ft) = s\, \theta_r(g \otimes f)\, t$.

(2) Prove that θ_r is bijective.

(3) Let $p : V \to V'$ be right S-linear and $q : W \to W'$ a bimodule map. Prove $(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*)$.

(4) Let $S = \Bbbk^{\oplus 3}$ with idempotents e_1, e_2, e_3, $V = \Bbbk a$ with $e_1 a e_2 = a$, and $W = \Bbbk c$ with $e_2 c e_3 = c$. Compute $W^* \otimes_S V^*$, $V^* \otimes_S W^*$ and $(V \otimes_S W)^*$, and describe θ_r. What happens if the two duals are tensored in the original order?

(5) Suppose instead that $S = \Bbbk[x]/(x^2)$ and $V = W = S/(x)$, with the same definitions of U^* and θ_r. Show that $W^* \otimes_S V^*$ and $(V \otimes_S W)^*$ are both 1-dimensional but $\theta_r = 0$. At which step of your proof of (2) does the argument fail?

Exercise 4. Quadratic presentations and Hilbert series
The parts are independent.

(1) Let $n \geq 3$, $Z \subset \{1, \dots, n-2\}$, and $A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$ with the quotient grading.
   i. Show that $\{e_{ii}\} \cup \{e_{ij} \mid i < j, \text{ no } k \in Z \text{ with } i \leq k \leq j-2\}$ is a basis of A_Z.
   ii. Write $A_Z = T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.
   iii. For $n = 5$ and $Z = \{1, 3\}$ compute $h_{A_Z}(t)$.
   iv. For $n = 4$ and $Z = \{1, 2\}$, show $e_{14} \in (e_{13}, e_{24})$, that $(A_Z)_2 = 0$, and that $A_Z = S \oplus V$ is radical-square-zero; compute $h_{A_Z}(t)$.

(2) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and let $V = \Bbbk a \oplus \Bbbk b$ be an S-bimodule with $e_1 a e_2 = a$, $e_2 b e_1 = b$ and all other idempotent actions zero. Find a basis of $T_S(V)_d$ for all $d \geq 0$ and compute $h_{T_S(V)}(t)$. Then find a basis and h for $T_S(V)/(ab)$. Which entries of the two matrices differ, and why is $T_S(V)/(ab)$ finite-dimensional?

   Hint. Count alternating words in each degree, including degree 0.

(3) Let V, W have bases $x_1, \dots, x_n$ and $y_1, \dots, y_m$, generators in degree 1, and consider $\mathsf{S}(V)$ and $\bigwedge(W)$
   i. Show that
   $$\dim_{\Bbbk} \mathsf{S}(V)_2 = \tfrac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\textstyle\bigwedge W)_2 = \tfrac{m(m-1)}{2}.$$
   ii. Using that ordered monomials form a basis of $\mathsf{S}(V)$ and strictly increasing products a basis of $\bigwedge W$, compute $\dim \mathsf{S}(V)_d$ and $\dim(\bigwedge W)_d$ for all d, and $h_{\mathsf{S}(V)}(t)$, $h_{\bigwedge W}(t)$.
   iii. Let $P = \Bbbk\langle x, y\rangle/(xy)$. Show that $\{y^a x^b \mid a, b \geq 0\}$ is a basis, and that $h_P(t) = h_{\Bbbk[x,y]}(t)$. Prove that P and $\Bbbk[x, y]$ are not isomorphic as algebras.

Exercise 5. Computing quadratic duals
The parts are independent.

(1) Let $A = \Bbbk A_4/(e_{13})$. Write a presentation of $A^!$ using reversed generators, and find homogeneous bases and the Hilbert series of A and $A^!$. Verify
$$h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I_4,$$
and check that dualising twice recovers A , identifying each twice-reversed generator with its original matrix unit.

   Hint. Among $e_{12}e_{23}$ and $e_{23}e_{34}$ only the first is a relation. Complement the composable pairs and reverse their order.

(2) Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1 with $C_{ij} = \dim_{\Bbbk} e_i V e_j$, and let $A = T_S(V)/(V \otimes_S V) = S \oplus V$.
   i. Prove $A^! = T_S(V^*)$, $h_A(t) = I + tC$, and $h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}$.
   ii. Deduce $h_A(t) h_{A^!}(-t)^{\mathsf{T}} = I$.
   iii. Specialise to $S = \Bbbk e_1 \oplus \Bbbk e_2$, $V = \Bbbk a \oplus \Bbbk b$, $e_1 a e_2 = a$, $e_2 b e_1 = b$, and write down both series.

   Hint. For (i) use $V \otimes_S V$ as the relation space and the formula $h_{T_S(V)}(t) = (I - tC)^{-1}$ of Lecture 2.

(3) For $q \in \Bbbk$ let $C_q = \Bbbk\langle x, y\rangle/(x^2, y^2, xy + qyx)$, $\deg x = \deg y = 1$.
   i. Find a homogeneous basis of C_q and compute all products of two basis elements and $h_{C_q}(t)$, treating $q = 0$ and $q \neq 0$. Justify linear independence of your basis.
   ii. For $u = \alpha x + \beta y$ compute u^2. For each q, determine the (α, β) with $u^2 = 0$, and the q for which every degree-1 element squares to zero.

iii. Prove: $C_q \cong C_{q'}$ as graded algebras if and only if either $q = q'$ or $qq' = 1$. Deduce that $C_1 \ncong C_{-1}$ when $\operatorname{char} \Bbbk \neq 2$ (they differ already as ungraded algebras, since one is commutative), and say what changes in characteristic 2.

iv. Compute a quadratic presentation of $C_q^!$ using θ_r, and its Hilbert series. Verify $h_{C_q}(t)h_{C_q^!}(-t) = 1$. Compare $C_0^!$ with $C_1^!$.

Hint for (3). A graded isomorphism is determined by a linear bijection of degree-1 parts and preserves the set of lines $\Bbbk u$ with $u^2 = 0$. For (4), evaluate a general combination of X^2, XY, YX, Y^2 on x^2, y^2, and $xy + qyx$, with $X = x^*$, $Y = y^*$.

Exercise 6. Bases, Hilbert series, and quadratic duals
The parts are independent.

(1) **A square with a parameter.** Let $S = \Bbbk e_1 \oplus \Bbbk e_2 \oplus \Bbbk e_3 \oplus \Bbbk e_4$ and let V be an S-bimodule with basis a, b, c, d and

$$e_1 a e_2 = a, \quad e_2 b e_4 = b, \quad e_1 c e_3 = c, \quad e_3 d e_4 = d,$$

and all other idempotent actions zero. The only composable words of length 2 are ab and cd. For $\lambda \in \Bbbk$ set $A_\lambda = T_S(V)/(ab - \lambda cd)$.

i. For every λ, including 0, find a presentation of $A_\lambda^!$, bases of A_λ and $A_\lambda^!$, and both Hilbert series. Do they depend on λ? Dualise again to recover the relation.

ii. Prove that $A_\lambda \cong A_\mu$ as graded algebras over S if and only if $\lambda = \mu = 0$ or both are non-zero.

iii. Show $A_1^! \cong (A_1)^{\mathrm{op}}$ as graded algebras over S.

Hint for (1). Evaluate $\mu b^* a^* + \nu d^* c^*$ on $ab - \lambda cd$. For (2), a graded isomorphism over S sends each of a, b, c, d to a scalar multiple of itself (or swaps the two paths); for (3) rescale d^*.

(2) **Counting words.** Let $B = T_\Bbbk(\Bbbk x \oplus \Bbbk y)/(x^2)$, $\deg x = \deg y = 1$.

i. Describe a basis of B_d, and compute $a_d = \dim B_d$ for $0 \le d \le 6$.

ii. Express a_d through a_{d-1}, a_{d-2} by sorting words by first letter, and find $h_B(t)$ as a rational function.

iii. Compute $B^!$ (with $X = x^*$, $Y = y^*$), a homogeneous basis, and $h_{B^!}(t)$. Verify $h_B(t) h_{B^!}(-t) = 1$.